Discrete and Continuous Methods for Packing Problems

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Contents

Preface

This book develops a unified framework for packing problems, combining discrete optimization, graph theory, and continuous methods from applied mathematics.

Chapter 1

Optimal Packing of Non-Convex Polygons

This project is an excercise in optimization and High performance computing. It is inspired in the classical problem of sphere packing with a twist: We seek to find the smallest square container that can fit N identical non-convex polygons without overlap.

1.1 Mathematical Formulation

We consider a set of N identical non-convex polygons

$$= \{P_1, \ldots, P_N\},$$

Each polygon can be described with the state variables $(x_i, y_i, \theta_i) \in \mathbb{R}^2 \times [0, 2\pi)$.

Definition 1.1 (Feasible Configuration)**.** A configuration is feasible if:

1. All vertices lie within the square domain $[0, L]^2$.
2. No two polygons overlap.

To objective is to find the minimal L such that all polygons are placed inside the square without overlap. Formally, we seek to solve:

$$\begin{aligned} \min_{L,} \quad & L \\ \text{s.t.} \quad & \text{is a feasible configuration of in } [0, L]^2. \end{aligned}$$

1.1.1 Feasibility-Based Search

A common strategy is to reduce the optimization problem to a sequence of feasibility tests. For a fixed container size L, one asks whether there exists a configuration such that all polygons fit inside $[0, L]^2$ without overlap. This leads to the decision problem

$$\text{Find} \quad \text{such that} \;\; \text{is feasible in } [0, L]^2.$$

By performing a search over values of L. Using bisection, one can approximate the minimal feasible container size. This approach shifts the difficulty from direct optimization to repeated feasibility detection in a highly constrained configuration space.

1.1.2 Energy-Based Formulations

An alternative and widely used approach replaces hard geometric constraints with soft penalties by introducing an energy function of the form

$$E(; L) = E_{\text{overlap}}() + E_{\text{boundary}}(; L),$$

where overlaps and boundary violations contribute positive energy. In this formulation, feasible configurations correspond to zero-energy states.

This transforms the constrained optimization problem into an unconstrained (but non-convex) minimization problem over $\mathbb{R}^{2N} \times S^1$. The resulting energy landscape is typically rugged, containing many local minima corresponding to jammed or partially overlapping configurations.

Energy-based formulations naturally motivate the use of stochastic optimization techniques, such as simulated annealing and Monte Carlo methods. We use them to explore the configuration space and find feasible or near-feasible configurations. which are designed to explore complex non-convex landscapes.

Computational and algorithmic challenges. The packing problem studied in this work presents several fundamental difficulties, both geometric and algorithmic.

1. **Highly non-convex energy landscape.** The objective function associated with packing design by associating a penalties to overlaps and boundary violations is highly non-convex. Additionally, it exhibits a large number of local minima. As will be shown in the next chapter, even simplified model like the Ising model possesses a non-convex energy landscape. As a consequence, purely local optimization methods get trapped in suboptimal configurations, requiring use of global or stochastic optimization techniques.

2. **Geometric complexity due to non-convex shapes.** By the nature of non-convex polygons, the overlap checks require careful geometric computation, often involving decomposition into convex components or the evaluation of separating axes. This geometric complexity significantly increases the computational cost of evaluating candidate configurations.

3. **Exponential growth of the search space.** The configuration space grows exponentially with the number of objects N, as each polygon contributes translational and rotational degrees of freedom. Consequently, exhaustive search methods become infeasible even for moderately large systems.

4. **Computational intractability (NP-hardness).** The problem is computationally intractable in general. This follows from the simplified version being hard: if all polygons are restricted to be circles. This can be seen by considering the associated *decision problem*: given a set of circles with prescribed radii and a square container of fixed side length, determine whether all circles can be placed inside the square without overlap.

   This decision problem is known as the *circle packing problem* and has been shown to be NP-complete. Since circle packing is a special case of the more general polygon packing problem, it follows that the general problem is NP-hard.

Chapter 2

Simulated Annealing and Its Parallel Variants

The energy-based formulation of Chapter 1 reduces polygon packing to minimizing a highly non-convex function $E(;L)$ over $\mathcal{C} = \mathbb{R}^{2N} \times [0, 2\pi)^N$. This chapter develops the stochastic optimization machinery used to explore that landscape. We begin by establishing the common probabilistic target, then derive the base Markov chain dynamics and the three parallel coordination strategies—MS, ER-MS, and PT—as distinct approximations to sampling from that target.

2.1 The Common Probabilistic Target

All three methods studied in this work can be understood as strategies for approximating samples from the *Gibbs measure* at temperature $T > 0$:

$$\pi_T() \;=\; \frac{1}{Z(T)} \exp\left(-\frac{E(;L)}{T}\right), \qquad Z(T) = \int_{\mathcal{C}} e^{-E(;L)/T}\, d.$$

As $T \to 0$, π_T concentrates on the global minimizers of E. Feasible configurations correspond to zero-energy states, so finding a feasible packing is equivalent to sampling from π_T in the low-temperature limit.

The methods differ in how they allocate R parallel workers to approximate this target:

- **MS** draws R independent samples from separate realizations of the annealing process.
- **ER-MS** couples workers via a periodic selection–mutation operator, forming an interacting particle system that biases mass toward low-energy regions.
- **PT** couples R chains thermodynamically via swap moves, targeting the *joint* Gibbs distribution over the replica ensemble.

2.2 Base Markov Chain Dynamics

2.2.1 Metropolis–Hastings Kernel

Given a configuration $_k$ at step k, a candidate $'$ is drawn from a symmetric Gaussian proposal kernel $q(\,\cdot\,|_k)$ and accepted with probability

$$\alpha(_k, ') \;=\; \min\Big(1,\; e^{-\Delta E/T_k}\Big), \qquad \Delta E = E('; L) - E(_k; L).$$

Symmetry of q ensures this satisfies detailed balance with respect to π_{T_k}. The resulting transition kernel is

$$P_k(\to d') = q('|)\,\alpha(, ')\,d' + r()\,\mathbf{1}(d'),$$

where $r() = 1 - \int q('|)\,\alpha(, ')\,d'$ is the rejection probability.

Proposal structure. At each step, a polygon index $i \sim \text{Uniform}\{1, \ldots, N\}$ is selected and a joint positional and rotational perturbation is applied:

$$x_i' = x_i + \delta_x, \quad y_i' = y_i + \delta_y, \quad \theta_i' = \theta_i + \delta_\theta,$$

with $(\delta_x, \delta_y)^\top \sim \mathcal{N}(\mathbf{0}, \sigma_{\text{pos}}^2 I_2)$ and $\delta_\theta \sim \mathcal{N}(0, \sigma_\theta^2)$, drawn independently.

2.2.2 Cooling Schedule and Mixing

We use a *geometric* cooling schedule:

$$T_k = T_{\max}\,\rho^k, \qquad \rho = \left(\frac{T_{\min}}{T_{\max}}\right)^{1/K},$$

where K is the total number of steps in a bisection probe. The geometric schedule is characterized by a constant multiplicative temperature reduction ρ per step, which gives equal thermal exposure at each decade of energy: the number of steps spent in any interval $[cT, T]$ is $-\log c/\log(1/\rho)$, independent of the absolute temperature level.

The theoretically sufficient condition for convergence to the global minimum requires logarithmic cooling $T_k = c/\log(1+k)$ for a constant c exceeding the depth of the deepest local energy barrier [**?**]. Such schedules are computationally infeasible for the problem sizes considered here; the geometric schedule instead prioritizes practical convergence within a fixed computational budget, with $T_{\min}$ and $T_{\max}$ fixed by pilot calibration (Section **??**).

2.2.3 Integration with Bisection

Each method operates within an outer bisection loop over the container side length L. For a fixed bracket $[L_{\text{lo}}, L_{\text{hi}}]$, a probe evaluates the midpoint $L = (L_{\text{lo}} + L_{\text{hi}})/2$ by running K steps of the SA-based search. If a feasible configuration is found, the bracket contracts: $L_{\mathrm{hi}} \leftarrow L$; otherwise $L_{\mathrm{lo}} \leftarrow L$. The SA energy function $E(;L)$ is therefore re-evaluated at a different L on each bisection step, and the Gibbs target π_T changes accordingly.

2.3 Method A: Multi-Start Simulated Annealing (MS)

2.3.1 Definition

MS instantiates R independent Markov chains, each evolving under its own copy of the base SA kernel:

$$ {}^{(r)}_{k} \xrightarrow{P_k, T_k} {}^{(r)}_{k+1}, \qquad r = 1, \ldots, R. $$

Workers share no state during a probe. Seeds are generated deterministically as

$$ s_r = \mathrm{splitmix64}\big(g \oplus \mathrm{hash}(N, \texttt{run\_id}, \texttt{probe\_idx}, r)\big), $$

where g is the global seed, ensuring exact reproducibility.

2.3.2 Probabilistic Analysis

Let p denote the probability that a single SA chain finds a feasible configuration within the allotted probe time τ. The probability that at least one of R independent workers succeeds is

$$ P_{\mathrm{success}} = 1 - (1 - p)^R. $$

For small p, this grows as Rp to first order, so R workers provide a linear improvement in success probability at linear cost. More precisely, if the first-success time for a single chain is Geometric(p)-distributed with mean $1/p$, then the minimum across R independent chains has mean $1/(Rp)$: the expected time to first feasibility scales as $1/R$.

2.3.3 Limitations

The independence of MS chains is both its computational strength (zero communication overhead) and its fundamental weakness. Workers cannot exploit configurations discovered by their siblings, leading to redundant exploration of the same basins. In highly multimodal landscapes with narrow feasible regions, this redundancy becomes the dominant cost.

2.4 Method B: Elite-Resample Multi-Start (ER-MS)

2.4.1 Motivation and Interpretation

ER-MS augments MS by introducing a periodic selection–mutation operator that concentrates the worker population toward low-energy regions. Viewed abstractly, the R workers form an *interacting particle system* whose empirical distribution

$$\mu_k^R = \frac{1}{R} \sum_{r=1}^{R} \delta_{{}^{(r)}_{k}}$$

is driven toward π_T by alternating between independent SA evolution and population-level reweighting. This mirrors the structure of Sequential Monte Carlo (SMC) samplers, where a population of weighted particles is evolved and periodically resampled.

2.4.2 Resampling Operator

A resampling event is triggered every $K_{\text{resample}} = 200N$ MH steps. Workers are ranked by energy and the elite pool is defined as the top

$$k' = \lceil 0.25R \rceil$$

workers. Elite workers are preserved unchanged. Each non-elite worker r is replaced by a perturbed copy of a uniformly drawn elite r^*:

$${}^{(r)} \leftarrow {}^{(r^*)} + \xi_r, \quad r^* \sim \text{Uniform}\{1, \dots, k'\}.$$

2.4.3 Perturbation Model

The perturbation ξ_r is drawn independently for spatial and angular components. For polygon i, the displacement and rotation noise are

$$(\delta x_i, \delta y_i)^\top \sim \mathcal{N}\Big(\mathbf{0},\, \sigma_{\text{pos}}(f)^2\, I_2\Big), \qquad \delta\theta_i \sim \mathcal{N}\Big(0,\, \sigma_\theta(f)^2\Big),$$

where the standard deviations decay linearly with the elapsed fraction $f \in [0, 1]$ of the current probe:

$$\sigma_{\text{pos}}(f) = (0.02 - 0.015\, f)\, L, \tag{2.1}$$

$$\sigma_\theta(f) = 5^\circ - 4^\circ f. \tag{2.2}$$

The full per-polygon perturbation covariance is therefore the block-diagonal matrix

$$\Sigma(f) = \mathrm{diag}\big(\underbrace{\sigma_{\mathrm{pos}}^2, \sigma_{\mathrm{pos}}^2, \sigma_\theta^2}_{\text{polygon } 1}, \; \dots, \; \underbrace{\sigma_{\mathrm{pos}}^2, \sigma_{\mathrm{pos}}^2, \sigma_\theta^2}_{\text{polygon } N}\big).$$

Large early-probe noise allows workers to escape elite basins and explore new regions; decaying late-probe noise concentrates refinement near the current best solution.

2.4.4 Anti-Collapse Mechanism

Population collapse—all workers converging to the same local minimum— reduces ER-MS to a single chain and eliminates the benefit of parallelism. Collapse is detected via the normalized energy spread

$$\delta_{\mathrm{rel}} = \frac{E_{\mathrm{max}} - E_{\mathrm{min}}}{\max(1, \, |E_{\mathrm{mean}}|)} < 10^{-6},$$

evaluated at each resampling event. If this condition holds for two consecutive resamples, the perturbation scales are doubled for the immediately following resample cycle:

$$\sigma_{\mathrm{pos}} \leftarrow 2\,\sigma_{\mathrm{pos}}(f), \qquad \sigma_\theta \leftarrow 2\,\sigma_\theta(f).$$

2.4.5 Relationship to SMC

In standard SMC, particles carry importance weights and resampling is performed proportional to those weights. ER-MS replaces importance weights with a hard rank cutoff (top k'), which is computationally cheaper but introduces selection bias: the retained configurations are not guaranteed to be representative samples from π_T. The mutation step (Gaussian perturbation) plays the role of the MCMC rejuvenation kernel in SMC, restoring diversity lost to selection.

2.5 Method C: Parallel Tempering (PT)

2.5.1 Replica Ensemble and Joint Distribution

PT maintains R replicas at a fixed geometric temperature ladder:

$$T_r = T_{\mathrm{min}} \left(\frac{T_{\mathrm{max}}}{T_{\mathrm{min}}} \right)^{(r-1)/(R-1)}, \qquad r = 1, \dots, R,$$

so that $T_1 < T_2 < \cdots < T_R$. The extended system targets the *joint* Gibbs distribution

$$\Pi\left({}^{(1)}, \dots, {}^{(R)}\right) = \prod_{r=1}^{R} \pi_{T_r}\left({}^{(r)}\right).$$

The geometric ladder spacing gives a constant temperature ratio between adjacent rungs, which empirically yields uniform swap acceptance rates across the ladder.

2.5.2 Within-Replica Dynamics

Between swap events, each replica evolves independently under its own MH kernel targeting π_{T_r}:

$$ {}^{(r)}_{k} \xrightarrow{P_k^{(r)},\,T_r} {}^{(r)}_{k+1}. $$

2.5.3 Replica Exchange

Every $K_{\text{swap}} = 200N$ MH steps, an exchange is proposed between each adjacent pair $(r, r+1)$. The swap acceptance probability is

$$\alpha_{\text{swap}}^{(r,r+1)} = \min\Bigg(1,\ \exp\Bigg[\left(\frac{1}{T_r} - \frac{1}{T_{r+1}}\right)\Big(E({}^{(r+1)}; L) - E({}^{(r)}; L)\Big)\Bigg]\Bigg).$$

Proposition 2.1 (Detailed Balance for Replica Exchange)**.** *The swap acceptance probability* $\alpha_{\text{swap}}^{(r,r+1)}$ *satisfies detailed balance with respect to the joint distribution* Π*.*

Proof. Let and $'$ be the ensemble states before and after swapping replicas r and $r+1$. The ratio of joint probabilities is

$$\frac{\Pi(')}{\Pi()} = \frac{\pi_{T_r}({}^{(r+1)})\,\pi_{T_{r+1}}({}^{(r)})}{\pi_{T_r}({}^{(r)})\,\pi_{T_{r+1}}({}^{(r+1)})} = \exp\Bigg[\left(\frac{1}{T_r} - \frac{1}{T_{r+1}}\right)\Big(E^{(r+1)} - E^{(r)}\Big)\Bigg],$$

where $E^{(r)} = E({}^{(r)}; L)$. The Metropolis acceptance ratio $\alpha_{\text{swap}}^{(r,r+1)} = \min(1, \Pi(')/\Pi())$ then satisfies $\Pi()\,\alpha_{\text{swap}}^{(r,r+1)} = \Pi(')\,\alpha_{\text{swap}}^{(r+1,r)}$ by the standard Metropolis argument. $\square$

2.5.4 Mechanism: Temperature-Assisted Basin Escape

High-temperature replicas flatten the effective energy landscape, allowing them to traverse barriers that would trap a cold chain. When a swap carries a favorable configuration from replica $r+1$ to replica r, that configuration begins cold-chain refinement from a point already near a low-energy basin. This *temperature-assisted diffusion* is the core reason PT outperforms MS on landscapes with high, narrow barriers separating deep basins.

2.5.5 Early Stopping Criterion

Because our goal is feasibility rather than equilibrium sampling, a probe terminates only when the *coldest* replica ($r = 1$) reaches $E({}^{(1)}; L) \le \varepsilon_{\text{feas}}$. Feasible configurations visiting higher-temperature replicas are not accepted as solutions unless they propagate to replica 1 via an accepted swap chain.

2.5.6 Pilot Calibration

The temperature bounds $T_{\min}$ and $T_{\max}$ are set by the following pilot procedure:

1. Run a 10-minute pilot at $N = 100$, $R = 10$ with an initial temperature ladder.
2. Compute the empirical swap acceptance rate $\hat{\alpha}_r$ at each adjacent pair $(r, r+1)$.
3. If $\hat{\alpha}_r \notin$ [0.15, 0.60] for any rung, multiply $T_{\max}$ or $T_{\min}$ by a factor of 2 and repeat.
4. Lock the calibrated values. Run a 5-minute sanity pilot at $N = 200$ to confirm validity at larger system size.

The target window $[0.15, 0.60]$ is a standard empirical criterion: below 0.15 the cold chain mixes too slowly; above 0.60 adjacent replicas are too similar for exchange to provide useful information flow.

2.6 Energy Landscape Interpretation

The relative performance of the three methods is determined by the structure of the energy landscape $E(\cdot; L)$. Three geometric features are particularly relevant:

- **Local minima.** Configurations * satisfying $\nabla E = 0$ with $E(^*) > 0$ are jammed states: partially overlapping packings with no locally improving move.
- **Energy barriers.** A barrier of height Δ between two basins requires the chain to traverse a region of energy at least Δ above the current state. Under the Metropolis rule, the probability of acceptance at temperature T scales as $e^{-\Delta/T}$, so barriers suppress transitions exponentially as $T \to 0$.
- **Basins of attraction.** Each local minimum is surrounded by a basin of configurations from which gradient-following dynamics converge to it. Wide, shallow basins are easily escaped at moderate temperatures; deep, narrow feasible basins require low temperatures to be refined once entered.

MS relies on independent discovery of distinct basins; it does not improve mixing within any single chain. ER-MS concentrates the worker population in the lowest-energy basins discovered so far, amplifying exploitation at the cost of diversity. PT enables direct inter-basin transitions by lifting configurations to temperatures where barriers are effectively invisible, then returning them to low temperature for refinement.

Table 2.1: Structural comparison of parallel SA variants.

Property	MS	ER-MS	PT
Probabilistic target	π_T (each chain)	Biased toward low E	$\prod_r \pi_{T_r}$
Inter-worker comm.	None	Periodic (resample)	Periodic (swap)
Comm. trigger	—	Every $200N$ steps	Every $200N$ steps
Temperature structure	Shared schedule	Shared schedule	Fixed ladder
Elite/cold bias	None	Top $\lceil 0.25R \rceil$	Cold replica only
Barrier crossing	Weak (single-chain)	Moderate (population spread)	Strong (PT diffusion)
Early stop condition	Any worker feasible	Any worker feasible	Cold replica feasible
HPC pattern	Embarrassingly parallel	Barrier + collective sort	Neighbor communication

2.7 Summary

Table **??** summarizes the key structural properties of the three strategies.

The three methods represent a progression from zero coupling (MS) through unidirectional information flow driven by energy ranking (ER-MS) to bidirectional thermodynamic coupling governed by detailed balance (PT). Chapter 3 defines the energy function $E(;L)$ explicitly and develops the geometric algorithms required to evaluate it.